# CmR116_bmr17022
# Largest-cluster review

Primary analysis: 11,000 models. Cluster 1 contributes 8,669 models to the second pass.

Fresh automatic result: 3 child clusters. Exploratory review covers every K from 2 to 20.

The complete supplied constructs and the frozen primary residue/pair mask define the child analysis.

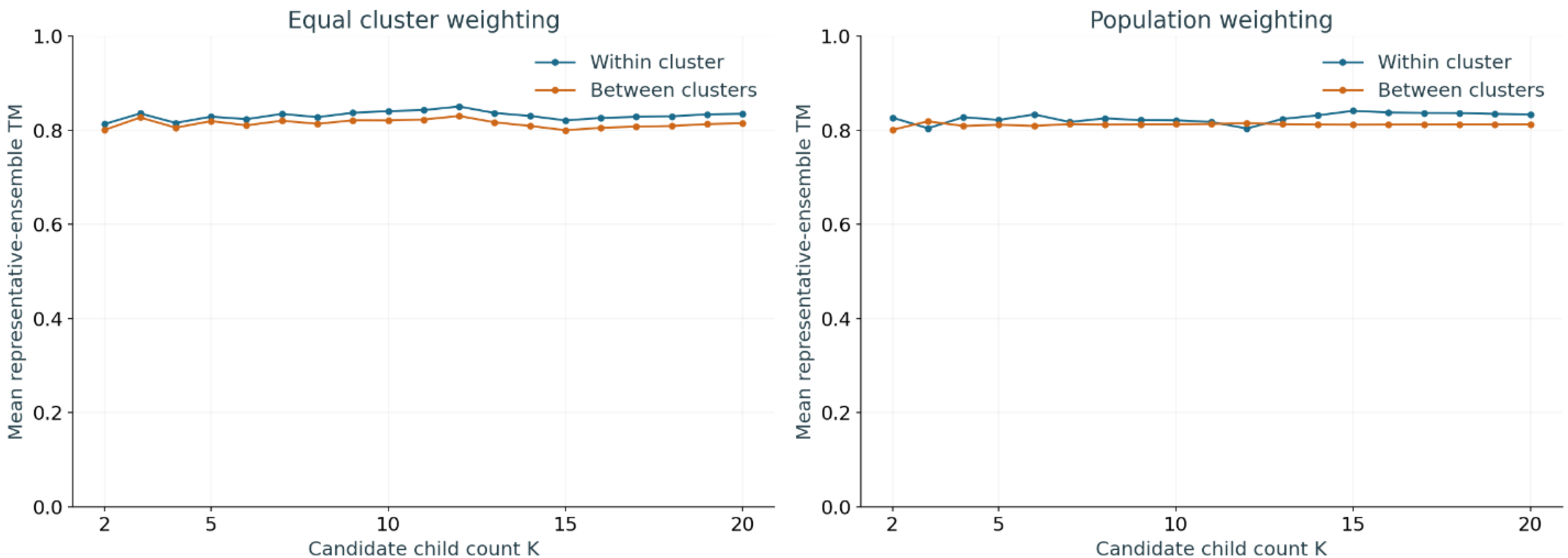
Fresh PCA, UMAP and t-SNE use only the selected parent-cluster models. Primary labels remain unchanged.

One fixed pool of up to 100 structures supplies all candidate ensemble-TM comparisons.

Within/between TM, cluster population and structural views support per-target review. Final K remains unselected.

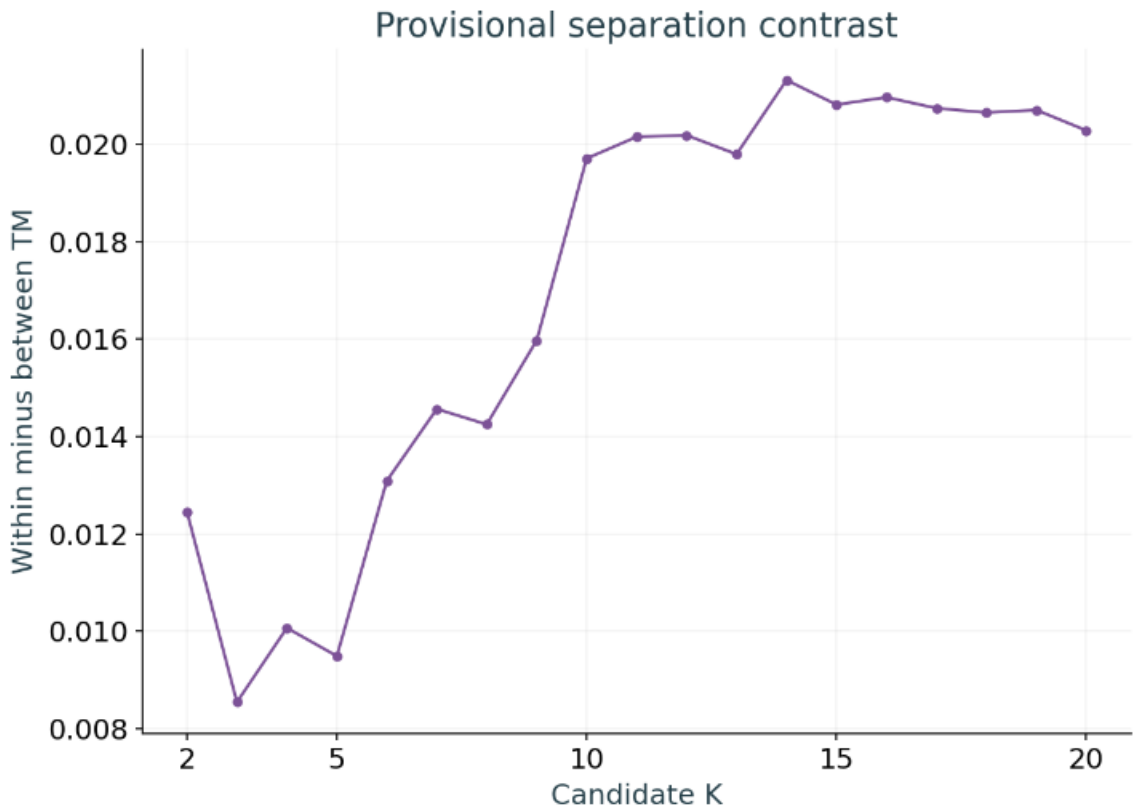
Two-level analysis. Child labels identify subsets of primary Cluster 1. The candidate cuts remain exploratory and await per-target review.

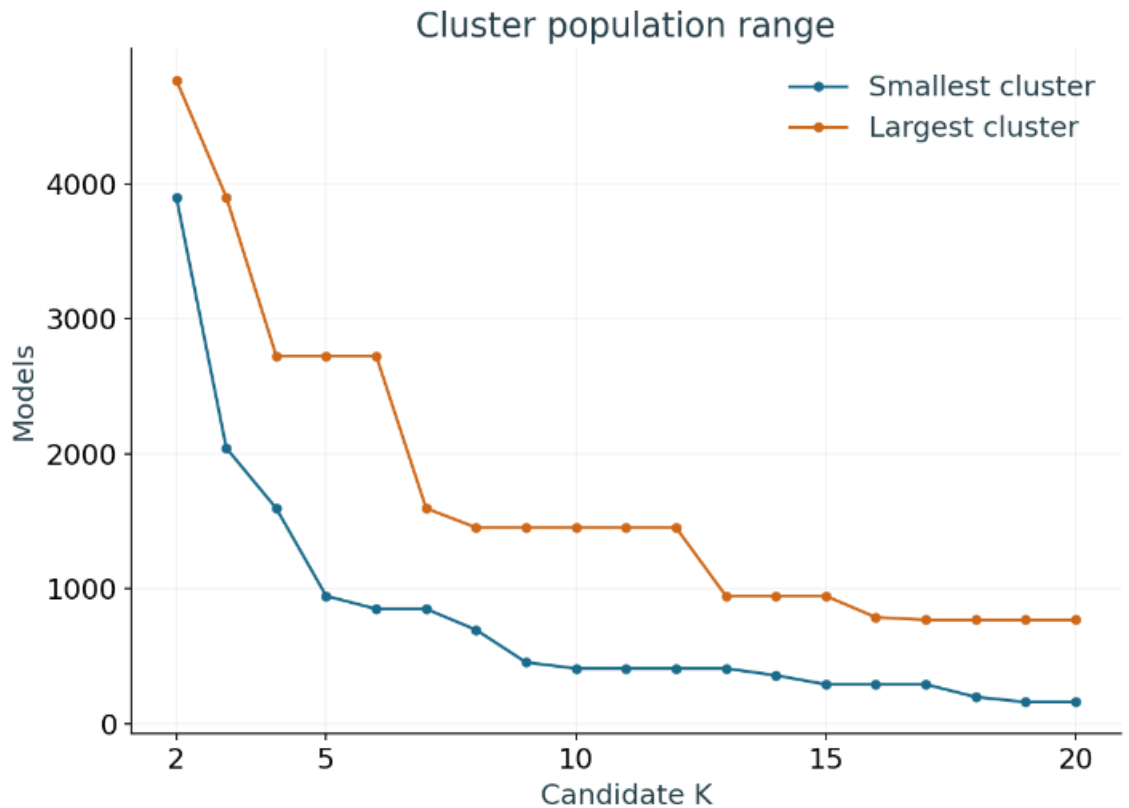
# CmR116_bmr17022 · Within/between ensemble similarity




All K values reuse the same fixed representative pool and TM matrix. Population weights reflect finest-group sizes. These are representative estimates, not all-model TM distributions.

# CmR116_bmr17022 · Count tradeoffs



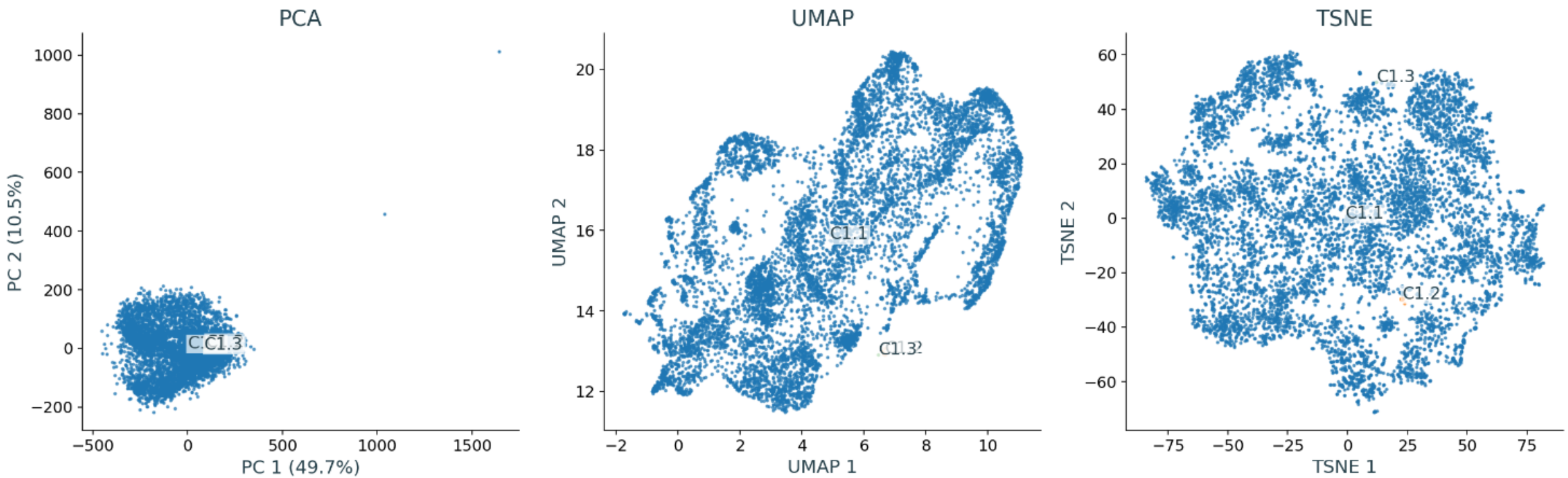




Contrast provides a provisional ranking. Increasing K can overpartition the ensemble, so this score does not determine an ideal state count. Review small groups and their structures.

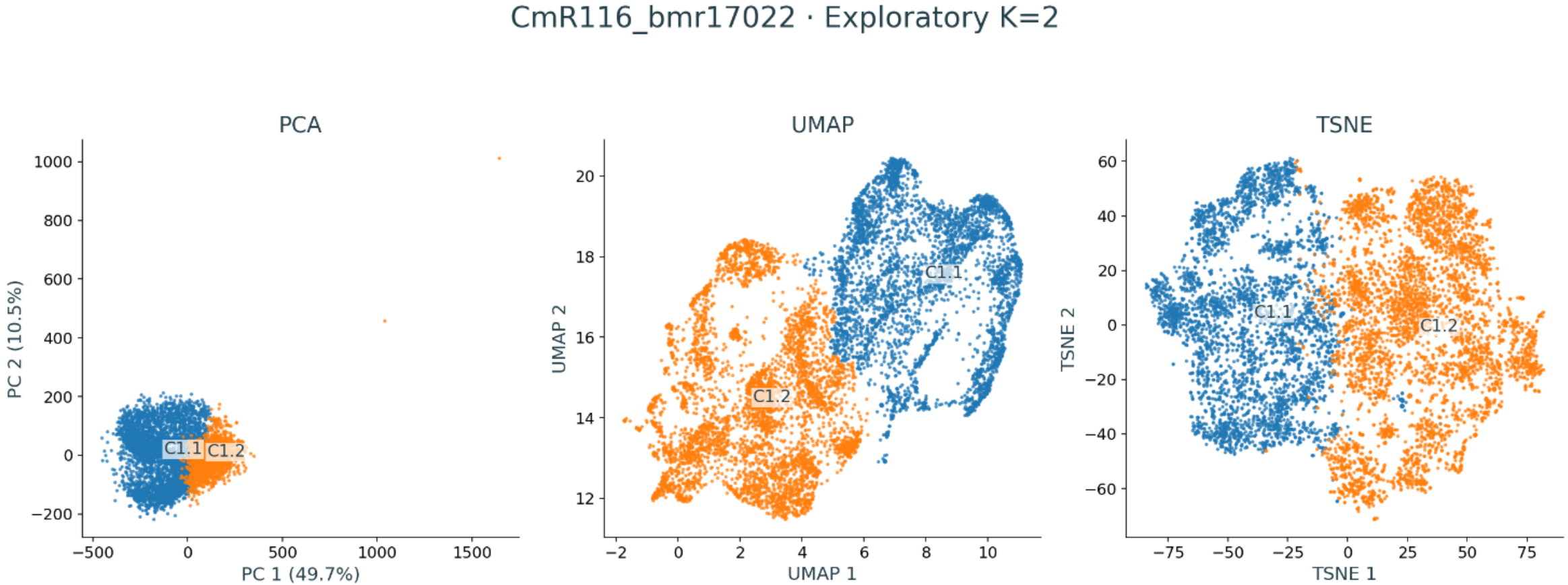
## CmR116_bmr17022 · Every candidate count

| K | Within TM | Between TM | Contrast | Min n | Max n | Singletons |
|---|---|---|---|---|---|---|
| 2 | 0.814 | 0.801 | 0.012 | 3,901 | 4,768 | 0 |
| 3 | 0.836 | 0.828 | 0.009 | 2,043 | 3,901 | 0 |
| 4 | 0.816 | 0.806 | 0.010 | 1,596 | 2,725 | 0 |
| 5 | 0.829 | 0.820 | 0.009 | 947 | 2,725 | 0 |
| 6 | 0.824 | 0.811 | 0.013 | 851 | 2,725 | 0 |
| 7 | 0.835 | 0.821 | 0.015 | 851 | 1,596 | 0 |
| 8 | 0.828 | 0.814 | 0.014 | 696 | 1,454 | 0 |
| 9 | 0.838 | 0.822 | 0.016 | 455 | 1,454 | 0 |
| 10 | 0.841 | 0.821 | 0.020 | 410 | 1,454 | 0 |
| 11 | 0.843 | 0.823 | 0.020 | 410 | 1,454 | 0 |
| 12 | 0.851 | 0.831 | 0.020 | 410 | 1,454 | 0 |
| 13 | 0.837 | 0.817 | 0.020 | 410 | 947 | 0 |
| 14 | 0.831 | 0.810 | 0.021 | 359 | 947 | 0 |
| 15 | 0.821 | 0.801 | 0.021 | 292 | 947 | 0 |
| 16 | 0.827 | 0.806 | 0.021 | 292 | 789 | 0 |
| 17 | 0.829 | 0.808 | 0.021 | 292 | 770 | 0 |
| 18 | 0.830 | 0.810 | 0.021 | 199 | 770 | 0 |
| 19 | 0.834 | 0.814 | 0.021 | 161 | 770 | 0 |
| 20 | 0.835 | 0.815 | 0.020 | 161 | 770 | 0 |

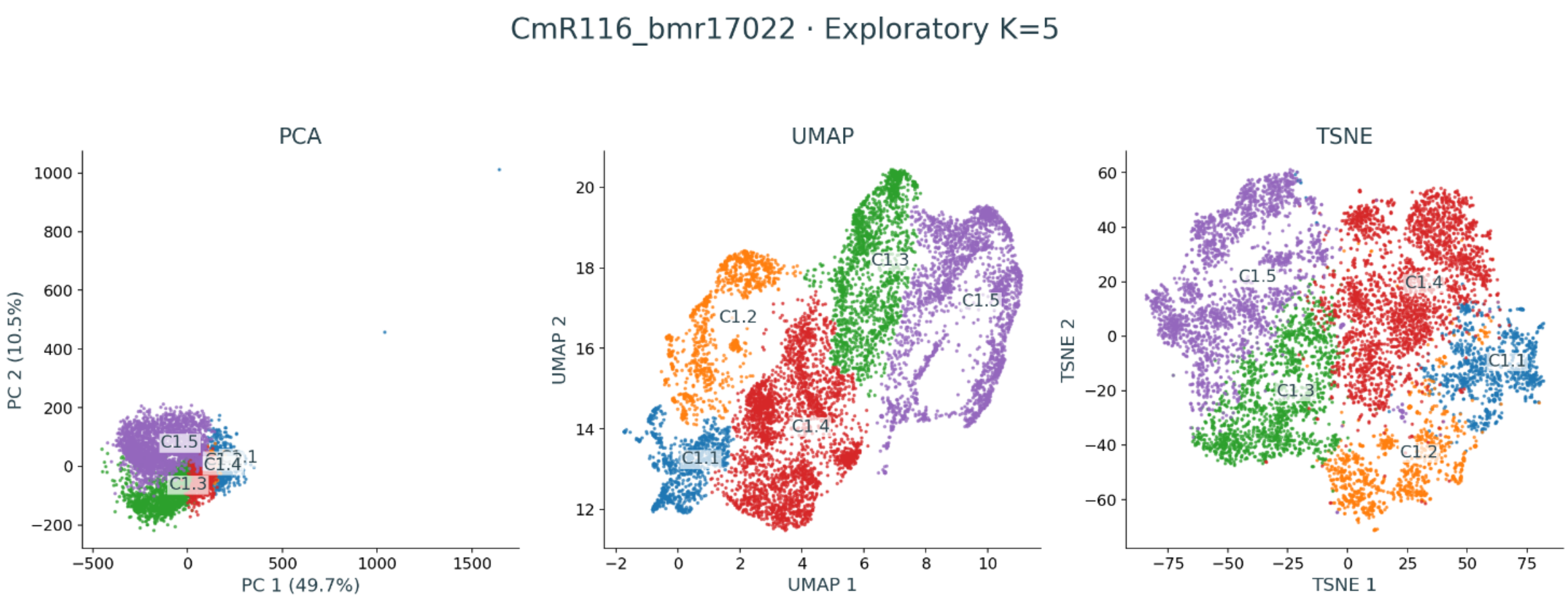
Equal-cluster TM estimates exclude self pairs. Unavailable within-cluster values occur when representative support is insufficient. The full CSV preserves every value and population count.



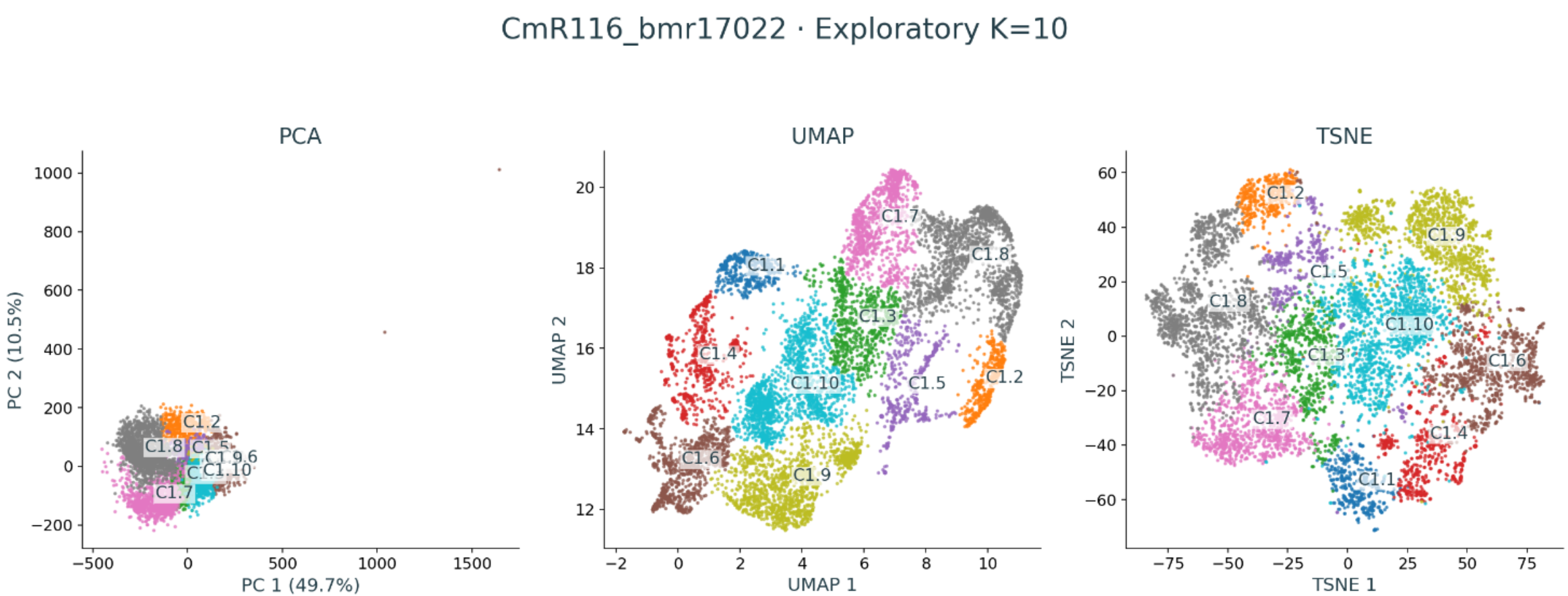

Automatic child baseline: 3 child clusters among 8,669 selected models. The three projections share these labels. All K=2–20 views are available in the interactive atlas.



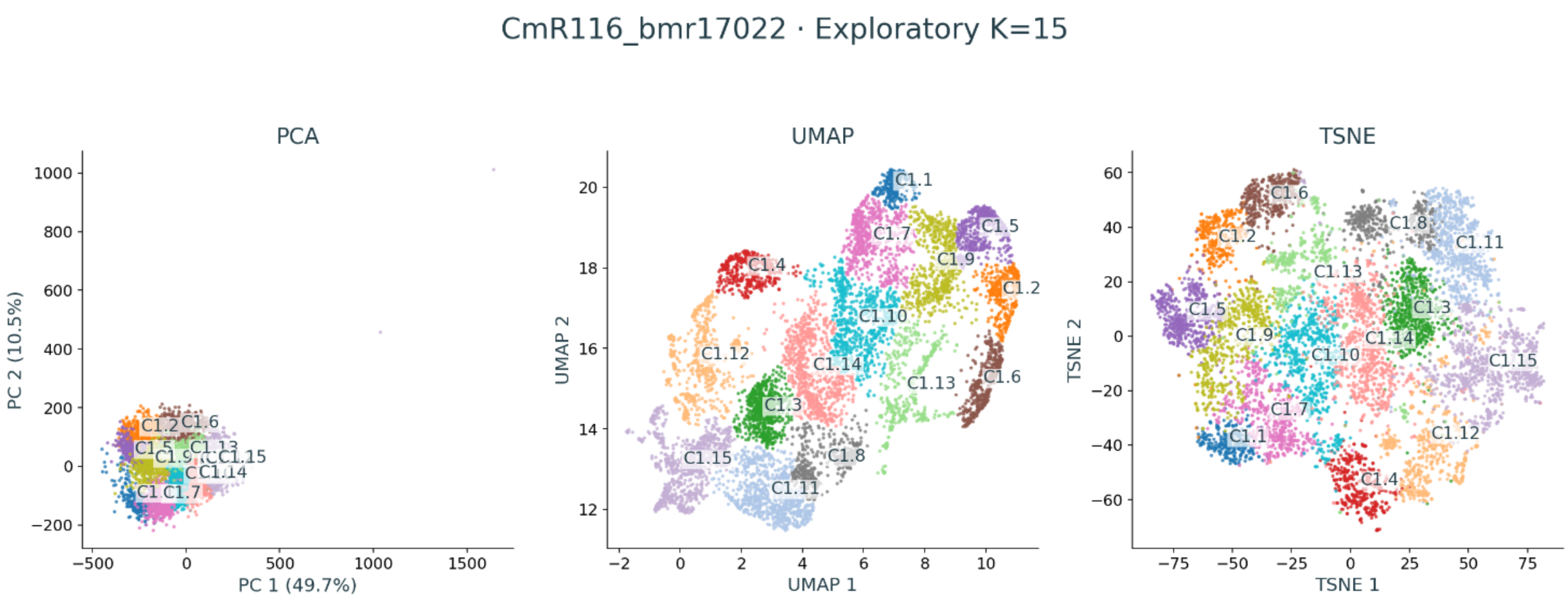

Exploratory K=2: 2 child clusters among 8,669 selected models. The three projections share these labels. All K=2–20 views are available in the interactive atlas.



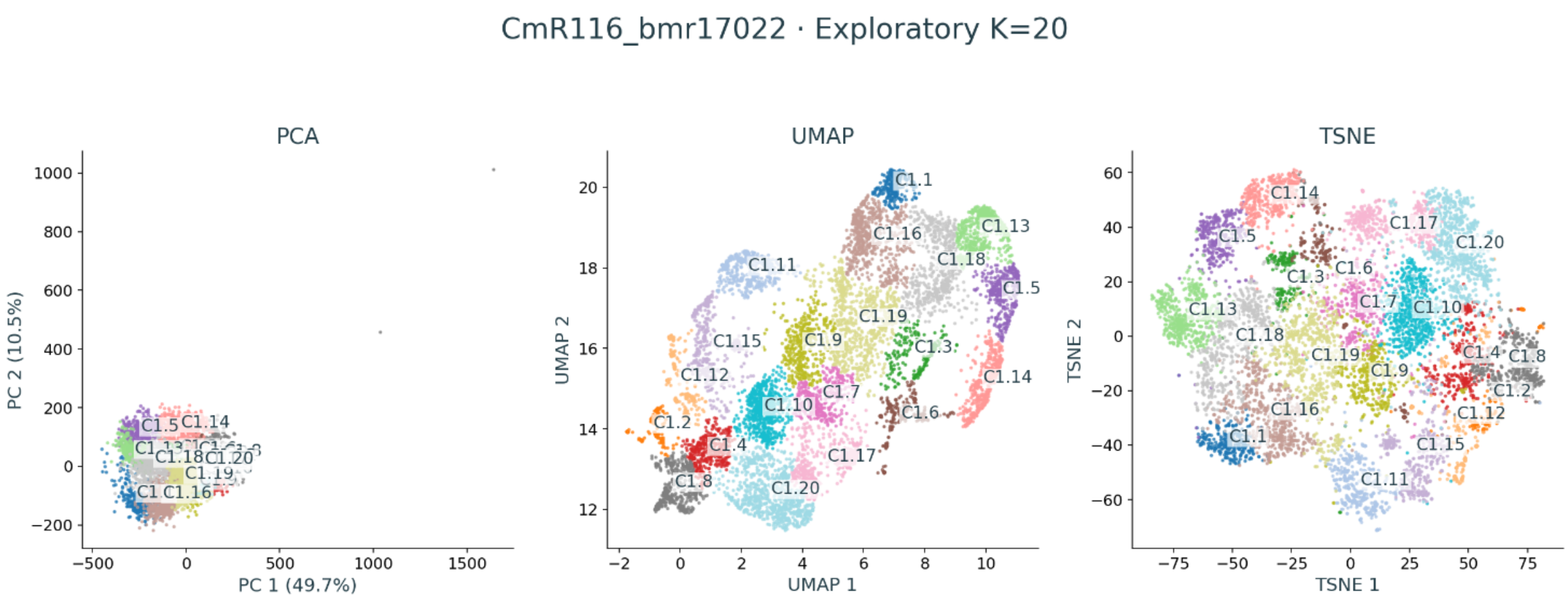

Exploratory K=5: 5 child clusters among 8,669 selected models. The three projections share these labels. All K=2–20 views are available in the interactive atlas.



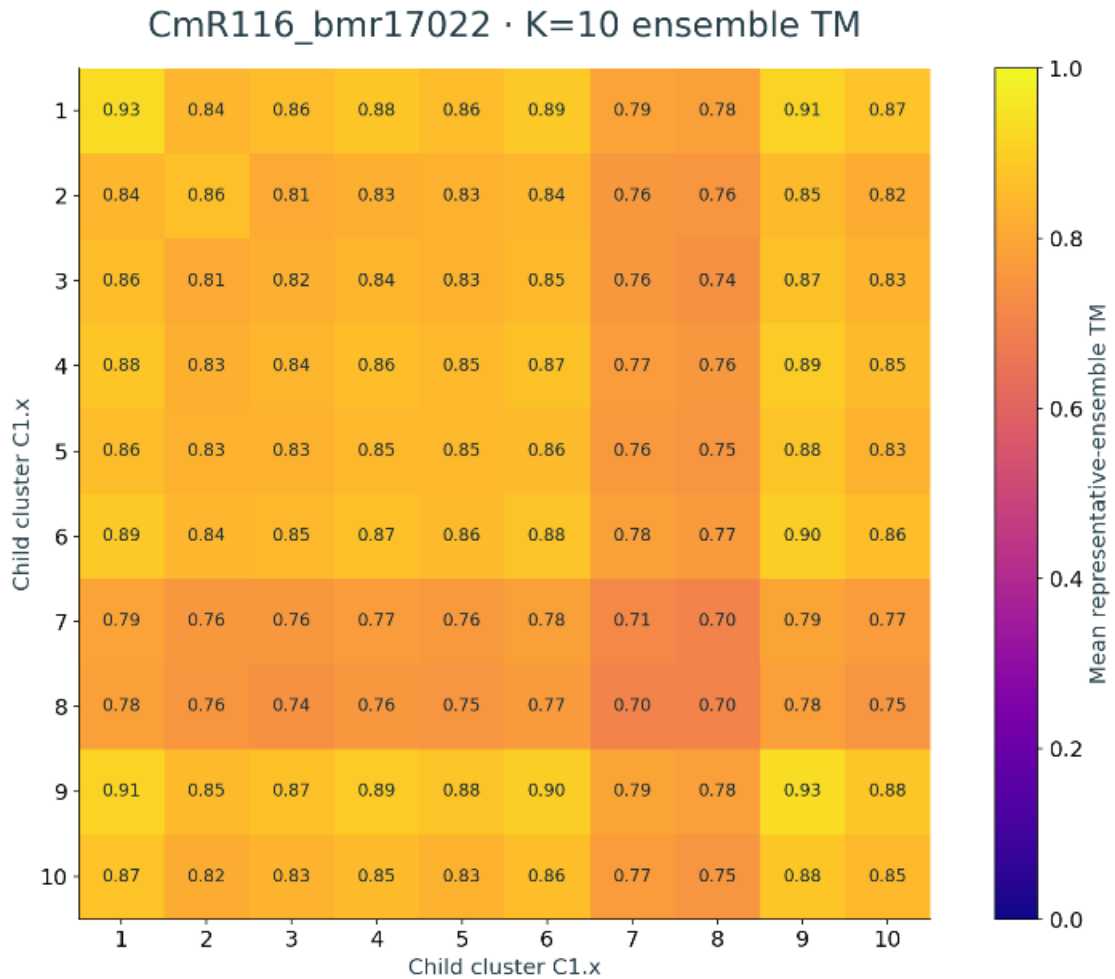

Exploratory K=10: 10 child clusters among 8,669 selected models. The three projections share these labels. All K=2–20 views are available in the interactive atlas.



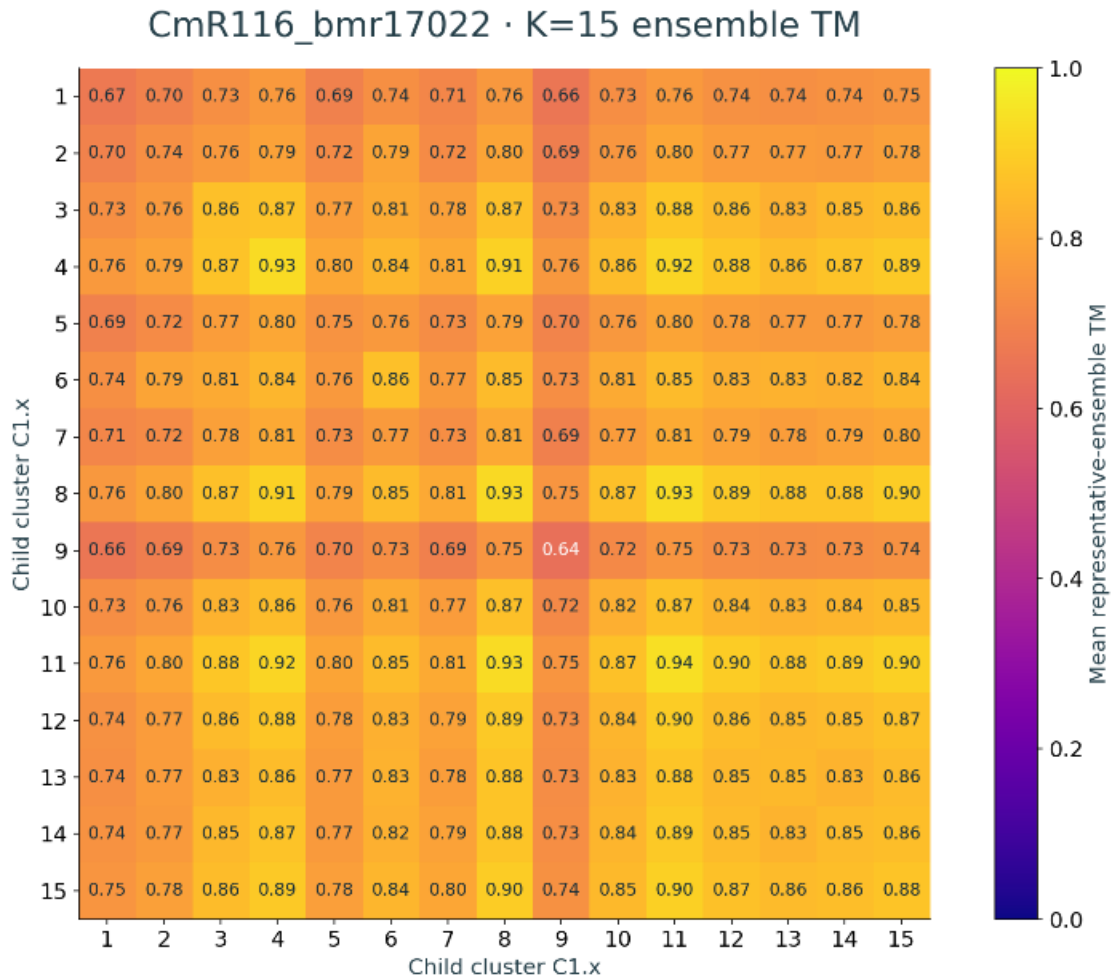

Exploratory K=15: 15 child clusters among 8,669 selected models. The three projections share these labels. All K=2–20 views are available in the interactive atlas.



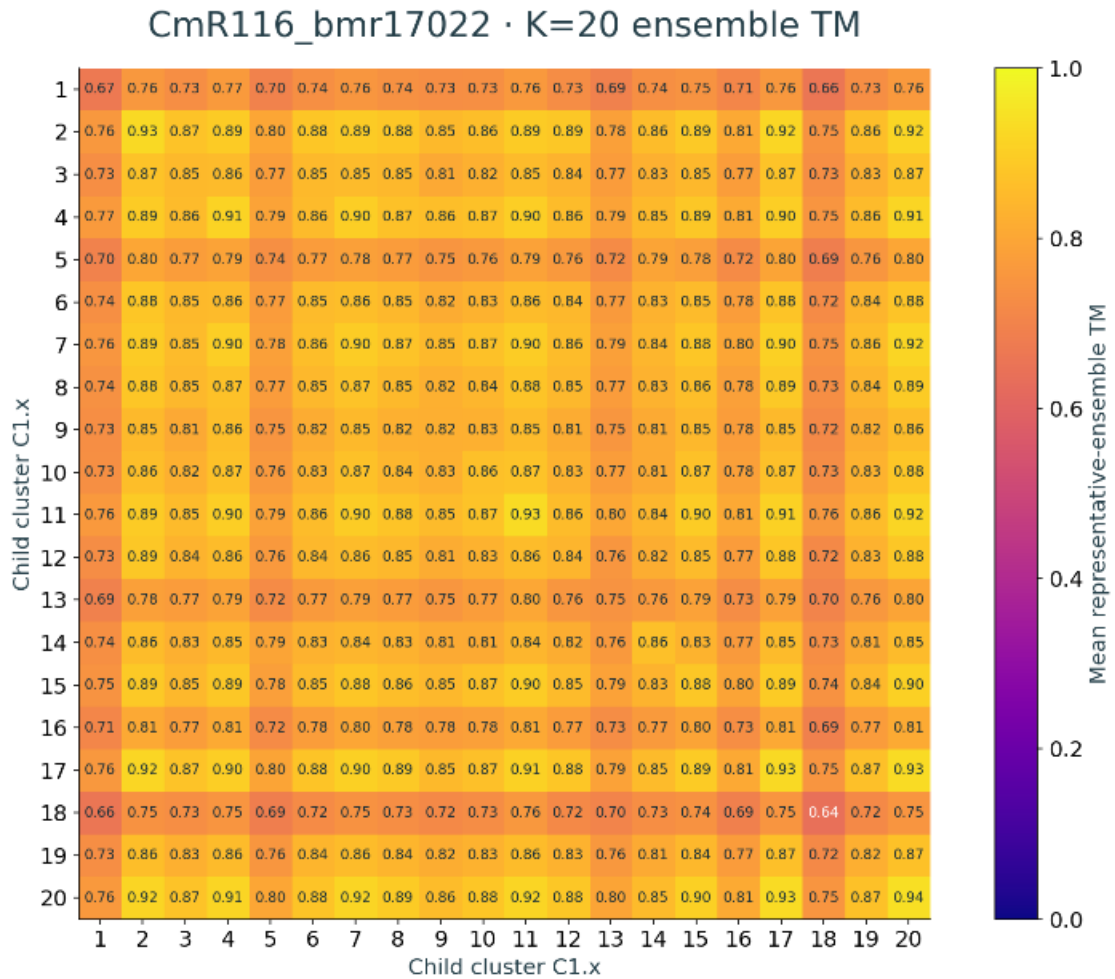

Exploratory K=20: 20 child clusters among 8,669 selected models. The three projections share these labels. All K=2–20 views are available in the interactive atlas.




Off-diagonal cells show weighted mean TM between representative ensembles. Diagonal cells show within-ensemble TM excluding self pairs. Blank estimates have insufficient representative support. The model pool is identical across K.




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